GLOBAL FORM OF THE NEARBY-OPEN QUASI-SECTION: A CURVE THROUGH X INSIDE U QUASI-FINITE OVER THE BASE AT X

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 14993-15004 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 15006-15023, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. This is the X-version cited-but-not-yet-opened when L16095 (change-hypotheses) was collected last wave; the citation chain in the lane is now closed in the right order. Sibling of L16081 (meeting-nearby-open without the X suffix): different hypothesis packaging, both counted per their distinct aggregate statements.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's local lemma quasi-finite-quasi-section-meeting-nearby-open (the affine-local algebraic version collected as an earlier wave card); elementary properties of local rings of schemes at a point and of reduced closures.

Source reference. Original lemma 05GT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14958–14967. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$ with image $s \in S$. Let $U \subset X$ be an open subscheme. Assume f locally of finite type, S locally Noetherian, x a closed point of X_s, and assume there exists a point $x' \in U$ with $x' \rightsquigarrow x$ and $f(x') \neq s$. Then there exists a closed subscheme $Z \subset X$ such that (a) $x \in Z$, (b) $f|_Z : Z \to S$ is quasi-finite at x, and (c) there exists a $z \in Z$, $z \in U$, $z \rightsquigarrow x$ and $f(z) \neq s$.*

Proof. This is a reformulation of Lemma 05GT. Namely, set $A = \mathcal{O}_{S,s}$ and $B = \mathcal{O}_{X,x}$. Denote $V \subset \mathrm{Spec}(B)$ the inverse image of U. The ring map $f^\sharp : A \to B$ is essentially of finite type. By assumption there exists at least one point of V which does not map to the closed point of $\mathrm{Spec}(A)$. Hence all the assumptions of Lemma 05GT hold and we obtain a prime $\mathfrak{q} \subset B$ which does not lie over $\mathfrak{m}_A$ and such that $A \to B/\mathfrak{q}$ is the localization of a quasi-finite ring map. Let $z \in X$ be the image of the point $\mathfrak{q}$ under the canonical morphism $\mathrm{Spec}(B) \to X$. Set $Z = \overline{\{z\}}$ with the

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induced reduced scheme structure. As $z \rightsquigarrow x$ we see that $x \in Z$ and $\mathcal{O}_{Z,x} = B/\mathfrak{q}$. By construction $Z \to S$ is quasi-finite at x. $\square$